

SHORT-TERM EFFECTS OF COMPLEX AND CONTRAST TRAINING IN SOCCER PLAYERS' VERTICAL JUMP, SPRINT, AND AGILITY ABILITIES

JOSÉ MANUEL VILAÇA MAIO ALVES,^{1,3} ANTÓNIO NATAL REBELO,² CATARINA ABRANTES,^{1,3}
AND JAIME SAMPAIO^{1,3}

¹*Sport Sciences Department, University of Trás-os-Montes e Alto Douro, Vila Real, Portugal;* ²*Faculty of Sport, University of Porto, Porto, Portugal;* and ³*Research Center in Sports Sciences, Health Sciences and Human Development, Vila Real, Portugal*

ABSTRACT

Alves, JMVM, Rebelo, AN, Abrantes, C, and Sampaio, J. Short-term effects of complex and contrast training in soccer players' vertical jump, sprint, and agility abilities. *J Strength Cond Res* 24(4): 936–941, 2010—The purpose of this study was to analyze the short-term effects of complex and contrast training (CCT) on vertical jump (squat and countermovement jump), sprint (5 and 15 m), and agility (505 Agility Test) abilities in soccer players. Twenty-three young elite Portuguese soccer players (age 17.4 ± 0.6 years) were divided into 2 experimental groups (G1, $n = 9$, and G2, $n = 8$) and 1 control group (G3, $n = 6$). Groups G1 and G2 have done their regular soccer training along with a 6-week strength training program of CCT, with 1 and 2 training sessions·wk⁻¹, respectively. G3 has been kept to their regular soccer training program. Each training session from the CCT program was organized in 3 stations in which a general exercise, a multiform exercise, and a specific exercise were performed. The load was increased by 5% from 1 repetition maximum each 2 weeks. Obtained results allowed identifying (a) a reduction in sprint times over 5 and 15 m (9.2 and 6.2% for G1 and 7.0 and 3.1%, for G2; $p < 0.05$) and (2) an increase on squat and jump (12.6% for G1 and 9.6% for G2; $p < 0.05$). The results suggested that the CCT induced the performance increase in 5 and 15 m sprint and in squat jump. Vertical jump and sprint performances after CCT program were not influenced by the number of CCT sessions per week (1 or 2 sessions·wk⁻¹). From the obtained results, it was suggested that the CCT is an adequate training strategy to develop soccer players' muscle power and speed.

KEY WORDS strength, muscle power, postactivation potentiation

Address correspondence to José Manuel Vilaça Maio Alves, vilaca9@gmail.com.

24(4)/936–941

Journal of Strength and Conditioning Research

© 2010 National Strength and Conditioning Association

936 ^{the}Journal of Strength and Conditioning Research

INTRODUCTION

Research concerning the impact of strength training in the performance of different sport skills have been showing both increases (15,17,28) and decreases (12,13,24) in performance. At this propose, Cometti (9) states that strength training programs must assure transference between the acquired strength and the main technical skills and presents specific guidelines for strength training based on complex and contrast training (CCT).

The complex training alternates biomechanical similar high-load weight training with plyometric exercises, set for set, in the same workout (4,7,16,28). This method is supported by the assumption of a postactivation potentiation (PAP) of the neuromuscular system (11,23). The PAP, as defined by Robbins (23), is a phenomenon by which the exerted muscle force is increased due to his previous contraction. The PAP effect in jump performance was studied by Gourgoulis et al. (17). The author identified an increase of 2.39% in vertical jump performance, when the jump was preceded by a squat exercise. It was also identified a higher increase in the jump height in athletes with greater maximum strength values. Nevertheless, Ebben et al. (13) failed to identify increases in force reaction average, in maximum reaction force, and in electromyography signal from the *pectoralis major* and *triceps brachii* muscles. These results were obtained after the fall of a medicine ball, carried out immediately after the horizontal bench press. The authors concluded that complex training is a good strategy to perform strength and plyometric training in the same session.

Contrast training consists in the use of high and low loads in the same strength training session (8,9,25). The loads used in contrast training can engage different regimens of contraction (8,9). Cometti (8) presented this method, consisting of accomplishing 6 repetition sets with loads between 60 and 80% of 1 repetition maximum (1RM), alternated with 6 repetition sets with loads between 30 and 50% of 1RM. The 2 types of sets are executed at maximum speed. This method is considered very efficient to increase power. In fact, several power training methods are being used

extensively with these high- and low-load intensity combinations (2,3,19).

Aagaard et al. (1) and Cronin et al. (10) stated that strength training must be integrated with sport skills training to improve skill performance. Although some studies pointed out complex training as a good method to increase sports skills power (15,16,18,28), others failed to identify this effect (13). According to Robbins (23), this strength training method should be more investigated to find more conclusive statements about its effectiveness. In fact, the literature regarding manipulation of training variables exploiting PAP, as expressed in the enhancement of athletic performance, is very scarce. The authors have investigated the acute effects of PAP over multiple sets (12,23). Therefore, results cannot be extrapolated to more chronic adaptations.

This training modality (CCT) could be compared with other modalities, aimed at developing power. With these comparisons, some conclusions could perhaps be drawn (e.g., efficacy of PAP with respect to chronic adaptation).

The aim of the present study was to identify the effect of CCT in the performance: (a) on the sprint at 5 and 15 m, (b) on squat (SJ) and countermovement jump (CMJ), and (c) on the agility (*505 Agility Test*). We also aimed to analyze the effect of the number of training sessions per week (1 vs. 2 sessions·wk⁻¹).

METHODS

Experimental Approach to the Problem

In this study, we aimed to identify the short-term effects of CCT on jump, sprint, and agility abilities of soccer players.

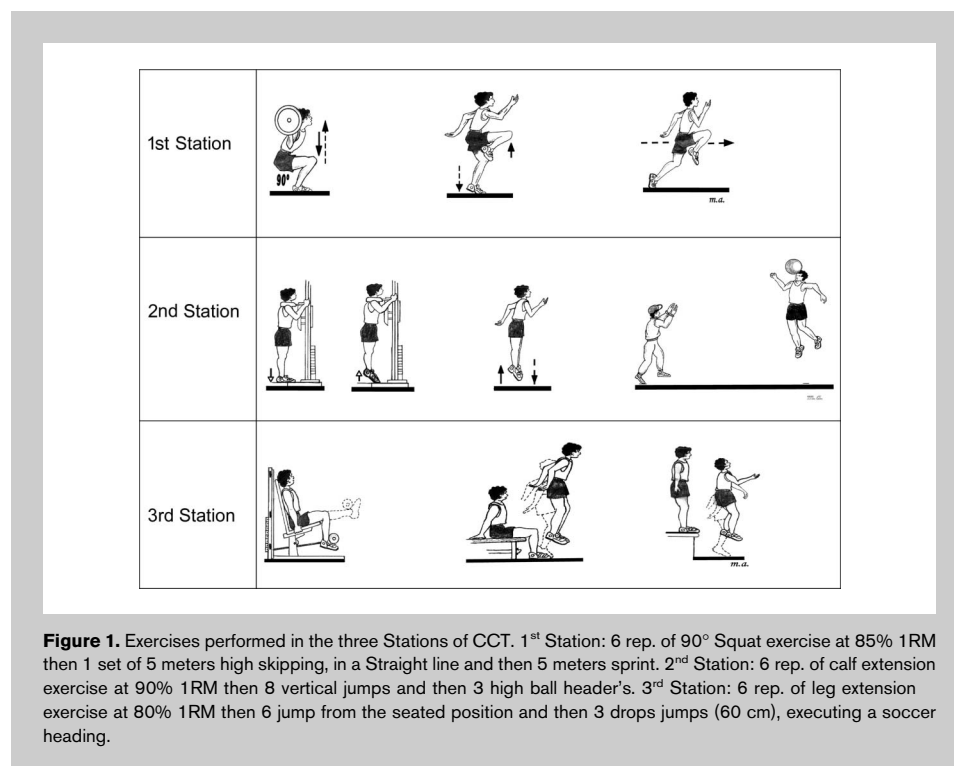
Players were divided into 3 groups (2 experimental groups and 1 control group). The 8-week preseason was divided in a period of 2 weeks for adaptation and a 6-week program of CCT applied in 2 experimental groups with 1 or 2 training sessions·wk⁻¹. Additionally, all groups performed their normal soccer training. The players were evaluated in the 5- and 15-m sprint (S5, S15), squat jump (SJ), countermovement jump (CMJ), and agility (AG), on 2 moments. The first evaluation was carried before the CCT program and the second after 6 weeks of CCT training.

Subjects

Twenty-three male soccer players (age: 17.4 ± 0.6 years; weight: 70.3 ± 8.3 kg; and height: 175.3 ± 6.3 cm) participating in the Portuguese elite championship were divided into 2 experimental (G1, *n* = 9 and G2, *n* = 8) and 1 control groups (G3, *n* = 6). All players and trainers were informed of the protocol, the experimental risks, and signed an informed consent document before the investigation. The informed consent form and the investigation were approved for use of Human Subjects by the Ethics Committee from the Sports Faculty in the University of Porto, Portugal.

Measurements

The determination of 1RM was carried through the procedures suggested by Kraemer and Fry (21). The evaluations were carried out after a period of 2 weeks, in which subjects learned the exercise execution techniques. The subjects were always kept under surveillance of professional technicians with high experience in strength training.



In the SJ, subjects performed a maximal vertical jump with hands on the waist, starting from an angle of 90° at the knee; in the countermovement jump, the subjects performed a maximal vertical jump starting from a standing position, with arm swing not allowed. All jumps were performed on the Ergojump (Globus Inc., Codogné, Italy) that recorded the flight time of all jumps. The flight time was used to calculate the change in the height of the body's center of gravity (Bosco et al., 1983). Subjects performed 3 trials in each protocol, and the best of them was used in the analysis.

Sprint evaluation was accomplished through a speed test that was carried out in a straight 15 m line (5). The times were measured through 3 pairs of

photoelectric cells (Speed Trap II – Browser Timing Systems) positioned at the starting point, at 5 and 15 m. Two attempts were granted being registered and considered the best one.

Agility was evaluated through the *505 Agility Test* (Ellis et al. (14)). This test aims at evaluating the capacity of the subjects to quickly change direction. Markers were set up at 5 and 15 m from a line marked on the ground. The subjects run from the 15-m marker toward the line (run in distance to build up speed) and through the 5 m markers, turned on the line, and run back through the 5 m markers. The time was recorded from when the subjects first run through the 5-m marker and stopped when they return through these markers (i.e., the time taken to cover the 5 m up and back distance—10 m total). The subjects were instructed to not overstep the line by too much, as this will increase their time. The subjects run 10 m in a straight line and touch with the foot (right or left) in a line placed at 5 m from this point, where they change direction and continue to run until crossing the starting point again. The time spent in the 10 m was measured by photoelectric cells (Speed Trap II; Browser Timing Systems, Draper, UT, USA) and considered for analysis. Each subject performed 3 attempts, and only the best one was considered. Reliability of all tests was high (intraclass correlation coefficient above 0.89).

Experimental Procedures

All subjects performed a 2-week adaptation strength training program with 3 sessions·wk⁻¹. The aim of this training was to provide the subjects an adaptation to strength training and to

the exercises used in the program. In this way, it was intended to optimize the exercise execution, prevent possible injuries, and attenuate the learning effect. The exercises were selected according to the muscle groups solicited in soccer, and the strength exercises were selected for the subsequent training program. In each exercise, 3 series of 12 repetitions were carried out. For the exercises soliciting abdominal and lumbar muscular groups, subjects performed 3 sets and, respectively, 30 and 20 repetitions. All training sessions began with a warm-up, consisting of 5 minutes of general exercises (low-intensity running, high skippings, leg flexions, lateral running, front and behind arm rotation, and sprints). Afterward, there was a 5-minute period of stretching exercises involving the muscle groups solicited in the training programs. Subjects were distributed for 9 exercise stations. The exercise load selected was of 60% of 1RM, except for abdominal and lumbar exercises, executed without additional external load. This training period lasted for 2 weeks (6 training sessions).

After finishing this adaptation period, all subjects were submitted to the first evaluation in the following tests: S5, S15; SJ; CMJ; and AG. After this evaluation, subjects were divided into 3 groups (2 experimental groups, G1 and G2 and 1 control group, G3). Subjects from G1 and G2 were evaluated in 1RM (squat, leg extension, and calf extension) aiming to determine the CCT workload.

Players from G1 and G2 performed their normal soccer training along with the CCT (1 session·wk⁻¹ to G1 and 2 sessions·wk⁻¹ to G2). The control group (G3) performed only the normal soccer training. The CCT was performed at the beginning of soccer practices (after the warm-up). Each training session was organized in 3 stations, in which a general exercise, a multiform exercise, and a specific exercise were performed. The exercises in each station were the following (Figure 1)—first station: the subjects performed a 85% of 1RM squat at 90°, repeated for 6 times; in continuation, 1 set of high skipping, cyclically, with the thighs parallel to the ground trying to keep a frequency of movement as high as possible during 5 m (in a straight line), was performed; this station was concluded with a straight line 5-m sprint. Second station: subjects performed a calf extension exercise, carried out 6 repetitions at 90% of 1RM; finished these exercise, players jumped vertically for

TABLE 1. Results (mean ± SD) of experimental groups (G1 and G2) and control group (G3) before and after CCT.*

Variables	G1	G2	G3
S5m (s)			
Pretest	1.09 ± 0.07	1.13 ± 0.04	1.13 ± 0.08
Posttest	0.99 ± 0.03‡	1.06 ± 0.03†	1.11 ± 0.03
S15m (s)			
Pretest	2.56 ± 0.10	2.57 ± 0.10	2.59 ± 0.07
Posttest	2.38 ± 0.09‡	2.49 ± 0.07†	2.56 ± 0.02
SJ (cm)			
Pretest	41.02 ± 6.11	39.68 ± 4.34	41.0 ± 3.05
Posttest	46.19 ± 7.70‡	43.50 ± 4.50†	40.70 ± 3.91
CMJ (cm)			
Pretest	42.84 ± 4.55	41.78 ± 5.28	42.63 ± 3.35
Posttest	42.92 ± 5.56	42.79 ± 4.45	41.53 ± 2.71
AG (s)			
Pretest	2.34 ± 0.11	2.32 ± 0.08	2.37 ± 0.09
Posttest	2.31 ± 0.09	2.32 ± 0.03	2.39 ± 0.15

*Sprint times at 5 m (S5 m) and at 15 m (S15 m), jump performance without countermovement (SJ), jump performance with countermovement (CMJ), time in the agility test (AG).

†p ≤ 0.05; ‡p ≤ 0.01 differences between trials.

TABLE 2. Post hoc comparison between studied groups.*

Variables	G1-G2	G1-G3	G2-G3
S5 m (s)	-0.7	-0.12 [†]	-0.05 [†]
S15 m (s)	-0.11	-0.18 [†]	-0.07
SJ (cm)	2.69	5.49 [†]	2.8 [†]
CMJ (cm)	0.13	1.39	1.26
AG (s)	-0.01	-0.08	-0.07

*Sprint times at 5 m (S5 m), and at 15 m (S15 m), jump performance without countermovement (SJ), jump performance with countermovement (CMJ), time in the agility test (AG).

[†] $p < 0.05$.

8 times, trying to minimize ground contact time, then performed 3 ball headers, jumping as high as possible. Third station: subjects performed the leg extension carrying out 6 repetitions at 80% of 1RM; jumped vertically for 6 times, trying to reach the highest point from the seated position on a stool; then, performed 3 drop jumps (60 cm) leaving for a vertical jump, executing a soccer heading, and trying to minimize ground contact time and maximize jump height. The load was increased at 5% from 1RM each 2 weeks. After finishing the CCT program, all subjects were reevaluated.

Statistical Analyses

For statistical analysis, a 3 (G1, G2, G3) $\times$ 2 (pretest, posttest) repeated measures analysis of variance (ANOVA) was carried out using group and trial as factors (between and within factors, respectively). A Tukey post hoc test was used to identify differences between groups and trials. All data undergoing ANOVA were tested for assumptions of normality, homogeneity of variance and covariance matrices, and sphericity. Neither assumption was violated. The level of significance was $p \leq 0.05$.

RESULTS

The results of the tests performed by the soccer players before and after CCT program are outlined in Table 1. After 6 weeks of CCT, the main results observed were as follows: a reduction of the sprint times over 5 and 15 m for G1 and G2 (9.17 and 6.19% for G1 and 7.03 and 3.11%, for G2; $p < 0.05$); an increase on SJ for both groups (12.6 for G1 and 9.63% for G2; $p < 0.05$); no significant differences in the CMJ and AG tests; and no significant differences in all tests between groups G1 and G2 (Table 2).

DISCUSSION

The aim of the present study was to identify the effects of a 6-week CCT program in soccer players' vertical jump, sprint, and agility abilities and to compare the effect of 1 vs. 2 training sessions·wk⁻¹.

After the CCT program, we found a reduction in sprint times over 5 and 15 m (G1 and G2) and an increase on SJ (G1 and G2). These data suggest that CCT can induce performance increases in 5- and 15-m sprint and in SJ.

Sprint results confirmed those obtained by Taïana et al. (26). These authors, applying a training program identical to the program used in the present study, identified reductions in 10 m (-0.08 seconds) and 30 m (-0.07 seconds) sprint times. In the present study, 5 and 15 m sprint times of G1 reduced 0.10 and 0.18 seconds, respectively. Likewise, Kotzamanidis et al. (20) identified a reduction of 0.25 seconds ($p \leq 0.05$) in soccer players 30 m sprint time, after the application of a strength training program combining loads between 3RM and 8RM with 4–6 sets of 30-m sprints. The training program used by Kotzamanidis et al. (20) included 2 training sessions·wk⁻¹ but was 3 weeks longer than our program. The results founded by the above referred authors and our results suggest that strength and complex training programs are useful practices to improve speed over distances between 5 and 30 m.

Subjects from G1 and G2 increased significantly SJ performances (G1: 5.17 cm and G2: 3.82 cm) after the application of the CCT. Kotzamanidis et al. (20) founded an increase of 1.99 cm ($p \leq 0.05$) in soccer players' SJ performances, after a strength training program with loads between 3RM and 8RM, combined with 4–6 sets of 30-m sprints. As mentioned before, the strength program analyzed by these authors lasts 9 weeks with a weekly training frequency of 2 sessions. The authors studied soccer players, who carried out only the strength training, and a control group, composed of subjects with similar age and moderate physical activity. Both groups have not improved the results on SJ test. These results reinforced the importance to combine strength training with exercises that make use of the stretch-shortening cycle, when it is intended to increase the vertical jump. Curiously, Taïana et al. (26) applied a program of training identical to ours and did not find any changes in SJ results. This fact could be explained by the high baseline performance of the soccer players participating in the study. In conclusion, the CCT seems to have favored increases in SJ, independently of the training session's frequency. The fact that the control group (G3) has not improved the SJ performance reinforces this idea.

We did not find any significant change in CMJ performance in any subject group after training. Taïana et al. (26) found a reduction of 3 cm in the jump height in the CMJ after the application of a strength training program identical to ours. The authors considered that these trends must have been related to the fact that the training program included only 1 weekly training session. According to the authors, improving jump performances would demand a minimum of 2 weekly training sessions. However, in the present study, the use of 2 training sessions·wk⁻¹ has not produced significant increases in CMJ jump height. Contrarily to the one observed in our study, Tricoli et al. (27) found a significant increase (2.8 cm)

in CMJ, in a group of athletes who used a strength training program that included exercises of Olympic weight combined with squat exercise. Another group of subjects was submitted to a strength training program combining jumps with squat exercises. This group showed a significant increase in the CMJ height (2.5 cm). This strength training program studied by Tricoli et al. (27) involved 8 weeks with 3 training sessions·wk⁻¹. This fact can lead to speculate that an insufficient weekly training frequency can justify the inefficiency of CCT to promote changes in CMJ performances.

Kotzamanidis et al. (20) also identified a significant increase in soccer players' CMJ performances. The authors used a strength training program with intensity loads between 3RM and 8RM, combined with 4–6 sets of 30-m sprints. This program included only 2 training sessions·wk⁻¹; however, when compared with the program of our study, the total duration was superior in 3 weeks. These data suggest that, besides the weekly frequency, the total training program duration can also influence the effectiveness of strength training programs.

Contrarily to what was verified in S5 and S15 tests, the AG test results from our study remained unchanged in all studied groups. One suggestion for these opposite results could be the fact that factors of agility and speed are fairly independent from one another (6,22,28,29). For this reason, AG training programs must be specific and independent from speed training programs (22,28,29). One second explanation for these results could be the fact that CCT does not have any exercise in which athletes had to perform changes in direction, breakings, and starts-movements, as demanded in the agility 505 Agility Test. Tricoli et al. (27) did not find, likewise, any increase of performance in the AG test used in their study. The authors suggested that the performance in movements that demand AG is more dependent from motor control factors than from maximum strength or muscular power.

PRACTICAL APPLICATIONS

Actual soccer competitive seasons played by young soccer players are becoming larger and more intense. Less time is remaining for preseasons; however, their importance in developing skills and promoting adequate conditioning increases. From the results obtained in the present study in elite young players, the 6-week strength training program (CCT) allowed to improve muscle power and speed. In fact, it is suggested that combining basic strength training with sport-specific movements may be an effective strategy to improve these skills in preseason.

REFERENCES

1. Aagaard, P, Simonsen, E, Trolle, M, Bangsbo, J, and Klausen, K. Specificity of training velocity and training load on gains in isokinetic knee joint strength. *Acta Physiol Scand* 156: 123–129, 1996.
2. Baker, D. Acute and long-term power responses to power training of an elite power athlete. *Strength Cond* 23: 47–56, 2001.
3. Baker, D. A series of studies on training of high-intensity muscle power in rugby league soccer players. *J Strength Cond Res* 15: 198–209, 2001.
4. Baker, D. Acute effect of alternating heavy and light resistances on power output during upper-body complex power training. *J Strength Cond Res* 17: 493–497, 2003.
5. Bosco, C, Luhtanen, P, and Komi, P. A simple method for measurement of mechanical power in jumping. *Eur J Appl Physiol* 50: 273–282, 1983.
6. Buttifant, D, Graham, K, and Cross, K. Agility and speed in soccer players are two different performance parameters. In: *Science and Football IV*. Spinks, W, Reilly, T, and Murphy, A, eds. Australia: Routledge, 1999. pp. 329–332.
7. Chiu, L, Fry, A, Weiss, L, Schilling, B, Brown, L, and Smith, S. Post activation potentiation responses in athletic and recreationally trained individuals. *J Strength Cond Res* 17: 671–677, 2003.
8. Cometti, G. *Los Métodos modernos de Musculación*. Editorial Paidotribo, Barcelona, Spain: 1998.
9. Cometti, G. *Fútbol y musculación*. INDE Publicaciones, Barcelona, Spain: 1999.
10. Cronin, J, McNair, P, and Marshall, R. Velocity specificity, combination training and sport specific tasks. *J Sci Med Sport* 4: 168–178, 2001.
11. Docherty, D, Robbins, D, and Hodgson, M. Complex training revisited: A review of its current status as a viable training approach. *Strength Cond* 26: 52–57, 2004.
12. Duthie, GM, Young, WB, and Aitken, DA. The acute effects of heavy loads on jump squat performances: An evaluation of the complex and contrast methods of power development. *J Strength Cond Res* 16: 530–538, 2002.
13. Ebben, WP, Jensen, R, and Blackard, DO. Electromyography and kinetic analysis of complex training variables. *J Strength Cond Res* 13: 451–456, 2000.
14. Ellis, L, Gastin, P, Lawrence, S, Savage, B, Buckeridge, A, Tumilty, D, Quinn, A, Woolford, S, and Young, W. *Protocols for the Physiological Assessments of Team Sports Players. Physiological Tests for Elite Athletes*. Australian Sports Commission, Champaign, IL: 2000. pp. 132.
15. Fatouros, IG, Jamurtas, AZ, Leontsini, D, Taxildaris, K, Aggelousis, N, Kostopoulos, N, and Buckenmeyer, P. Evaluation of plyometric exercise training, weight training and their combination on vertical jumping performance and leg strength. *J Strength Cond Res* 14: 470–476, 2000.
16. French, DN, Kraemer, WJ, and Cooke, CB. Changes in dynamic exercise performance following a sequence of preconditioning isometric muscle actions. *J Strength Cond Res* 17: 678–685, 2003.
17. Gourgoulis, V, Aggeloussis, N, Kasimatis, P, Mavromatis, G, and Garas, A. Effect of submaximal half squats warm-up program on vertical jump ability. *J Strength Cond Res* 17: 342–344, 2003.
18. Ingle, L, Sleep, M, and Tolfrey, K. The effect of a complex training and detraining programme on selected strength and power variables in early pubertal boys. *J Sport Sci* 24: 987–997, 2006.
19. Kawamori, N and Haff, GG. The optimal training load for the development of muscular power. *J Strength Cond Res* 18: 675–684, 2004.
20. Kotzamanidis, C, Chatzopoulos, D, Michailidis, C, Papaiakevou, G, and Patikas, D. The effect of combined high-intensity strength and speed training program on the running and jumping ability of soccer players. *J Strength Cond Res* 19: 369–375, 2005.
21. Kraemer, WJ, Ratamess, NA, Fry, AC, French, DN. Strength testing. Development and evaluation of methodology. In: *Physiological Assessment of Human Fitness*. Maud, J and Foster, C, eds. Human Kinetics, Champaign, IL: 1995. pp. 115–138.
22. Little, T and Williams, A. Specificity of acceleration, maximum speed, and agility in professional soccer players. *J Strength Cond Res* 19: 76–78, 2005.

23. Robbins, DW. Postactivation potentiation and its practical applicability: A brief review. *J Strength Cond Res* 19: 453–458, 2005.
24. Scott, SL and Docherty, D. Acute effects of heavy preloading on vertical and horizontal jump performance. *J Strength Cond Res* 18: 201–205, 2004.
25. Smilios, I, Pilianidis, T, Sotiropoulos, K, Antonakis, M, and Tokmakidis, SP. Short-term effects of selected exercise and load in contrast training on vertical jump performance. *J Strength Cond Res* 19: 135–139, 2005.
26. Taïana, F, Gréhaigne, J, and Cometti, G. The influence of maximal strength training of lower limbs of soccer players on their physical and kick performances. Science and Football II. In: *Proceedings of the Second World Congress of Science and Football*. Reilly, T, Clarys, J, and Stibbe, A, eds. Taylor and Francis, Eindhoven, the Netherlands: 1991.
27. Tricoli, V, Lamas, L, Carnevale, R, and Ugrinowitsch, C. Short-term effects on lower body functional power development: Weightlifting vs. vertical jump training programs. *J Strength Cond Res* 19: 433–437, 2005.
28. Young, WB, Jenner, A, and Griffiths, K. Acute enhancement of power performance from heavy load squats. *J Strength Cond Res* 12: 82–84, 1998.
29. Young, WB, McDowell, MH, and Scarlett, BJ. Specificity of sprint and agility training methods. *J Strength Cond Res* 15: 315–319, 2001.